Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/20/2005
PI Name:	Wang, Yu-li Ph.D.		
Project Title:	Effects of Gravity on the Structure and Function of Animal Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Immunology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	07/01/2001	**End Date:**	10/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG 2-1495		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Experiments will be performed with cells cultured on clear, flexible, protein-coated substrata, which mimic physiological environment and facilitate simultaneous mechanical stimulation and fluorescence imaging. Living fibroblasts will be probed with manipulations that simulate the effects of changes in gravity, such as pushing or pulling forces and changes in substrate flexibility or topography, and the responses will be analyzed by digital imaging of cytoskeletal structures, traction forces and signaling molecules. Conversely, cytoskeletal structures or signaling pathways will be manipulated with novel chemical, physical, or molecular biological means, and the impact on mechanosensing determined. Experiments will be focused on focal adhesions and microtubules, and on signaling pathways including calcium, small GTPases, and tyrosine phosphorylation. The general objective of this project is to determine how animal cells detect gravitational forces, and how gravitational forces affect cell migration. It is likely that many profound biological effects		

	associated with space flights are caused by reduced weight load and shifts in body fluid distribution, which in turn cause drastic changes in tissue mechanical properties and in mechanical input into cells. Thus to understand the physiological effects of gravity, it is crucial to determine how animal cells respond to such physical signals as substrate flexibility and mechanical forces. The proposed experiments will address these questions directly and will provide a solid foundation for understanding and treating health problems associated with space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cellular mechanosensing affects a number of physiological and pathological processes, from embryonic development, wounding healing, to cancer invasion. This project will provide important information for understanding these fundamental processes.
Task Progress:	We have made advances in two areas. First, we have used a model system developed in this project to analyze the mechanism of cellular mechanosensing. Fibroblasts are cultured within two layers of flexible, porous polyacrylamide substrates coated with extracellular matrix. We demonstrated previously that this environment allows engagement of both ventral and dorsal integrins, which leads to a dramatically elongated or branched morphology much like fibroblasts in tissues. We now found that calcium plays a key role in this process. Cells depleted of extracellular and intracellular calcium failed to respond to such engagement of dorsal integrin. Moreover, cells defective in calpain expression, or expressing a calpain-inhibiting peptide, also maintained a 2D spread morphology. We postulate that mechanical signals, mediated by integrin engagement, cause local entry of calcium, which in turn triggers intracellular calcium waves to suppress the formation of broad lamellipodia and cell spreading. Calpain, a calcium-activated protease localized at focal adhesions, represents one of possibly multiple effectors in driving the shape change. The second area of research is focused on cell behavior on substrates with topographic features. While most studies in cell biology are carried out with flat culture surfaces, it is known that most cells in the body make complex 3D contacts with the environment. We postulate that such complex pattern of contacts has a profound effect on cell behavior. Fibroblasts on polystyrene substrates with 15 micron pillars showed an elongated shape, with a concomitant decrease in the size and number of actin stress fibers and focal adhesions. Moreover, cell migration becomes highly erratic, apparently due to the increased adhesion at pillars. Interestingly, cells defective in focal adhesion kinase expression showed a reduced ability to respond to topographic features. These results, combined with previous results from this project, prompted us to propose that fibroblasts generally migrate toward maximal mechanical stimulations, which may be mediated through forces, rigidity, or topographic features.
Bibliography Type:	Description: (Last Updated: 06/16/2020)
Abstracts for Journals and Proceedings	Beningo K, Dembo M, Wang YL. "Vinculin is involved in the maturation and stabilization of focal adhesions." Presented at the 43rd American Society for Cell Biology Annual Meeting, San Francisco, CA, December 13-17, 2003. Mol Biol Cell. 2003 Dec;14:62a. , Dec-2003
Abstracts for Journals and Proceedings	Wang YL, Tsai I, Chua GC, Russell TP. "Area of substrate contact controls cell migration and cytoskeletal organization." Presented at the 43rd American Society for Cell Biology Annual Meeting, San Francisco, CA, December 13-17, 2003. Mol Biol Cell. 2003 Dec;14:325a. , Dec-2003
Articles in Peer-reviewed Journals	Munevar S, Wang YL, Dembo M. "Regulation of mechanical interactions between fibroblasts and the substratum by stretch-activated Ca2+ entry. " J Cell Sci. 2004 Jan 1;117(Pt 1):85-92. https://doi.org/10.1242/jcs.00795 ; Pubmed PMID: 14627625 , Jan-2004
Presentation	Wang. Yu-li. "Physical and Chemical Events at the Cell-Substratum Interface" Physics and Biology: A Materials Approach, Paris. Jun-2004